P2 Chapter 1: Numerical Methods

Chapter Practice

Key Points

1. If the function $f(x)$ is continuous on the interval $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, then $f(x)$ has at least one root, x, which satisfies $a < x < b$.
2. To solve an equation of the form $f(x) = 0$ by an iterative method, rearrange $f(x) = 0$ into the form $x = g(x)$ and use the iterative formula $x_{n+1} = g(x_n)$.
3. The Newton–Raphson formula for approximating the roots of a function $f(x)$ is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Chapter Exercises

1 $f(x) = x^3 - 6x - 2$

a Show that the equation $f(x) = 0$ can be written in the form $x = \pm\sqrt{a + \frac{b}{x}}$, and state the values of the integers a and b. **(2 marks)**

$f(x) = 0$ has one positive root, α.

The iterative formula $x_{n+1} = \sqrt{a + \frac{b}{x_n}}$, $x_0 = 2$ is used to find an approximate value for α.

b Calculate the values of x_1, x_2, x_3 and x_4 to 4 decimal places. **(3 marks)**

c By choosing a suitable interval, show that $\alpha = 2.602$ is correct to 3 decimal places. **(3 marks)**

2 $f(x) = \frac{1}{4 - x} + 3$

a Calculate $f(3.9)$ and $f(4.1)$. **(2 marks)**

b Explain why the equation $f(x) = 0$ does not have a root in the interval $3.9 < x < 4.1$. **(2 marks)**

The equation $f(x) = 0$ has a single root, α.

c Use algebra to find the exact value of α. **(2 marks)**

3 $p(x) = 4 - x^2$ and $q(x) = e^x$.

a On the same axes, sketch the curves of $y = p(x)$ and $y = q(x)$. **(2 marks)**

b State the number of positive roots and the number of negative roots of the equation $x^2 + e^x - 4 = 0$. **(1 mark)**

c Show that the equation $x^2 + e^x - 4 = 0$ can be written in the form $x = \pm(4 - e^x)^{\frac{1}{2}}$ **(2 marks)**

The iterative formula $x_{n+1} = -(4 - e^{x_n})^{\frac{1}{2}}$, $x_0 = -2$, is used to find an approximate value for the negative root.

d Calculate the values of x_1, x_2, x_3 and x_4 to 4 decimal places. **(3 marks)**

e Explain why the starting value $x_0 = 1.4$ will not produce a valid result with this formula. **(2 marks)**

Chapter Exercises

4 $g(x) = x^5 - 5x - 6$

a Show that $g(x) = 0$ has a root, α, between $x = 1$ and $x = 2$. **(2 marks)**

b Show that the equation $g(x) = 0$ can be written as $x = (px + q)^{\frac{1}{r}}$, where p, q and r are integers to be found. **(2 marks)**

The iterative formula $x_{n+1} = (px + q)^{\frac{1}{r}}$, $x_0 = 1$ is used to find an approximate value for α.

c Calculate the values of x_1, x_2 and x_3 to 4 decimal places. **(3 marks)**

d By choosing a suitable interval, show that $\alpha = 1.708$ is correct to 3 decimal places. **(3 marks)**

5 $g(x) = x^2 - 3x - 5$

a Show that the equation $g(x) = 0$ can be written as $x = \sqrt{3x + 5}$. **(1 mark)**

b Sketch on the same axes the graphs of $y = x$ and $y = \sqrt{3x + 5}$. **(2 marks)**

c Use your diagram to explain why the iterative formula $x_{n+1} = \sqrt{3x_n + 5}$ converges to a root of $g(x)$ when $x_0 = 1$. **(1 mark)**

$g(x) = 0$ can also be rearranged to form the iterative formula $x_{n+1} = \dfrac{x_n^2 - 5}{3}$

d With reference to a diagram, explain why this iterative formula diverges when $x_0 = 7$. **(3 marks)**

6 $f(x) = 5x - 4\sin x - 2$, where x is in radians.

a Show that $f(x) = 0$ has a root, α, between $x = 1.1$ and $x = 1.15$. **(2 marks)**

b Show that $f(x) = 0$ can be written as $x = p\sin x + q$, where p and q are rational numbers to be found. **(2 marks)**

c Starting with $x_0 = 1.1$, use the iterative formula $x_{n+1} = p\sin x_n + q$ with your values of p and q to calculate the values of x_1, x_2, x_3 and x_4 to 3 decimal places. **(3 marks)**

Chapter Exercises

7 **a** On the same axes, sketch the graphs of $y = \frac{1}{x}$ and $y = x + 3$. **(2 marks)**

b Write down the number of roots of the equation $\frac{1}{x} = x + 3$. **(1 mark)**

c Show that the positive root of the equation $\frac{1}{x} = x + 3$ lies in the interval $(0.30, 0.31)$. **(2 marks)**

d Show that the equation $\frac{1}{x} = x + 3$ may be written in the form $x^2 + 3x - 1 = 0$. **(2 marks)**

e Use the quadratic formula to find the positive root of the equation $x^2 + 3x - 1 = 0$ to 3 decimal places. **(2 marks)**

8 $g(x) = x^3 - 7x^2 + 2x + 4$

a Find $g'(x)$. **(2 marks)**

A root α of the equation $g(x) = 0$ lies in the interval $[6.5, 6.7]$.

b Taking 6.6 as a first approximation to α, apply the Newton–Raphson process once to $g(x)$ to obtain a second approximation to α. **(4 marks)**

c Given that $g(1) = 0$, find the exact value of the other two roots of $g(x)$. **(3 marks)**

d Calculate the percentage error of your answer in part **b**. **(2 marks)**

9 $f(x) = 2\sec x + 2x - 3$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$ where x is in radians.

a Show that $f(x) = 0$ has a solution, α, in the interval $0.4 < x < 0.5$. **(2 marks)**

b Taking 0.4 as a first approximation to α, apply the Newton–Raphson process once to $f(x)$ to obtain a second approximation to α. Give your answer to 3 decimal places. **(4 marks)**

c Show that $x = -1.190$ is a different solution, β, of $f(x) = 0$. **(2 marks)**

Chapter Exercises

10 $f(x) = e^{0.8x} - \dfrac{1}{3-2x}, \; x \neq \dfrac{3}{2}$

a Show that the equation $f(x) = 0$ can be written as $x = 1.5 - 0.5e^{-0.8x}$. **(3 marks)**

b Use the iterative formula $x_{n+1} = 1.5 - 0.5e^{-0.8x_n}$ with $x_0 = 1.3$ to obtain x_1, x_2 and x_3. Hence write down one root of $f(x) = 0$ correct to 3 decimal places. **(2 marks)**

c Show that the equation $f(x) = 0$ can be written in the form $x = p\ln(3 - 2x)$, stating the value of p. **(3 marks)**

d Use the iterative formula $x_{n+1} = p\ln(3 - 2x_n)$ with $x_0 = -2.6$ and the value of p found in part **c** to obtain x_1, x_2 and x_3. Hence write down a second root of $f(x) = 0$ correct to 2 decimal places. **(2 marks)**

11 **a** By writing $y = x^x$ in the form $\ln y = x\ln x$, show that $\dfrac{dy}{dx} = x^x(\ln x + 1)$. **(4 marks)**

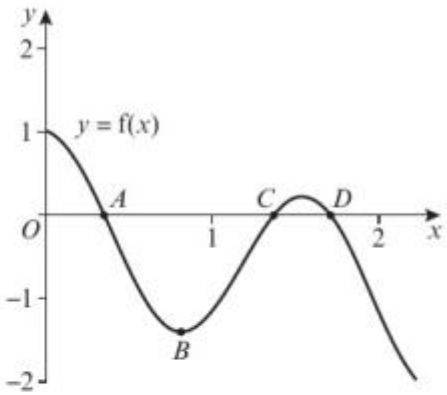
b Show that the function $f(x) = x^x - 2$ has a root, α, in the interval $[1.4, 1.6]$. **(2 marks)**

c Taking $x_0 = 1.5$ as a first approximation to α, apply the Newton–Raphson procedure once to obtain a second approximation to α, giving your answer to 4 decimal places. **(4 marks)**

d By considering a change of sign of $f(x)$ over a suitable interval, show that $\alpha = 1.5596$, correct to 4 decimal places. **(3 marks)**

Chapter Exercises

12 The diagram shows part of the curve with equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = \cos(4x) - \frac{1}{2}x$.



a Show that the curve has a root in the interval [1.3, 1.4]. **(2 marks)**

b Use differentiation to find the coordinates of point B. Write each coordinate correct to 3 decimal places. **(3 marks)**

c Using the iterative formula $x_{n+1} = \frac{1}{4}\arccos\left(\frac{1}{2}x_n\right)$, with $x_0 = 0.4$, find the values of x_1, x_2, x_3 and x_4. Give your answers to 4 decimal places. **(3 marks)**

d Using $x_0 = 1.7$ as a first approximation to the root at D, apply the Newton–Raphson procedure once to $\mathrm{f}(x)$ to find a second approximation to the root, giving your answer to 3 decimal places. **(4 marks)**

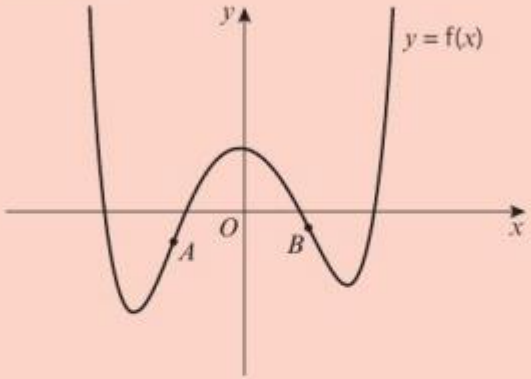
e By considering the change of sign of $\mathrm{f}(x)$ over an appropriate interval, show that the answer to part **d** is accurate to 3 decimal places. **(2 marks)**

Chapter Exercises

Challenge

$f(x) = x^6 + x^3 - 7x^2 - x + 3$

The diagram shows a sketch of $y = f(x)$. Points A and B are the points of inflection on the curve.



a Show that equation $f''(x) = 0$ can be written as:

i $x = \dfrac{7 - 15x^4}{3}$ **ii** $x = \dfrac{7}{15x^3 + 3}$ **iii** $x = \sqrt[4]{\dfrac{7 - 3x}{15}}$

b By choosing a suitable iterative formula and starting value, find an approximation for the x-coordinate of B, correct to 3 decimal places.

c Explain why you cannot use the same iterative formula to find an approximation for the x-coordinate of A.

d Use the Newton–Raphson method to find an estimate for the x-coordinate of A, correct to 3 decimal places.

Chapter Answers

1 **a** $x^3 - 6x - 2 = 0 \Rightarrow x^3 = 6x + 2$

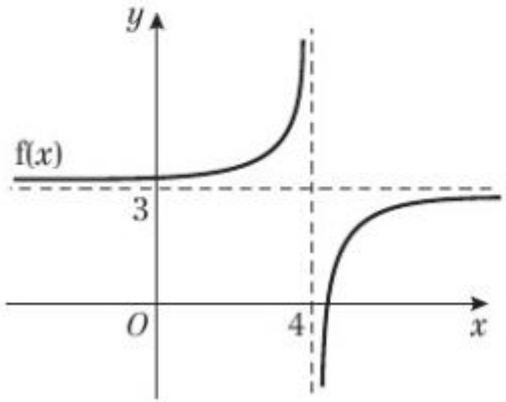
$\Rightarrow x^2 = 6 + \frac{2}{x} \Rightarrow x = \pm\sqrt{6 + \frac{2}{x}}$; $a = 6, b = 2$

b $x_1 = 2.6458, x_2 = 2.5992, x_3 = 2.6018, x_4 = 2.6017$

c $f(2.6015) = (2.6015)^3 - 6(2.6015) - 2 = -0.0025... < 0$
$f(2.6025) = (2.6025)^3 - 6(2.6025) - 2 = 0.0117 > 0$
There is a sign change in the interval $2.6015 < x < 2.6025$, so this implies there is a root in the interval.

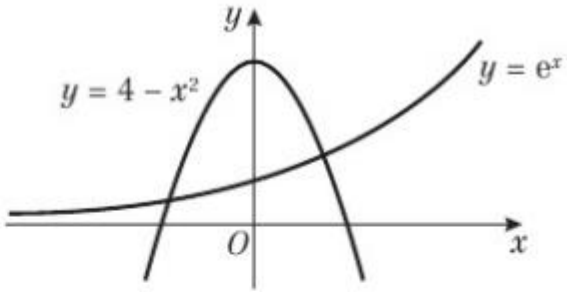
2 **a** $f(3.9) = 13$, $f(4.1) = -7$

b There is an asymptote at $x = 4$ which causes the change of sign, not a root.



c $\alpha = \frac{13}{3}$

3 **a**



b 2 roots – 1 positive and 1 negative

c $x^2 + e^x - 4 = 0 \Rightarrow x^2 = 4 - e^x \Rightarrow x = \pm(4 - e^x)^{1/2}$

d $x_1 = -1.9659$, $x_2 = -1.9647$, $x_3 = -1.9646$, $x_4 = -1.9646$

e You would need to take the square root of a negative number.

4 **a** $g(1) = -10 < 0$, $g(2) = 16 > 0$. The sign change implies there is a root in this interval.

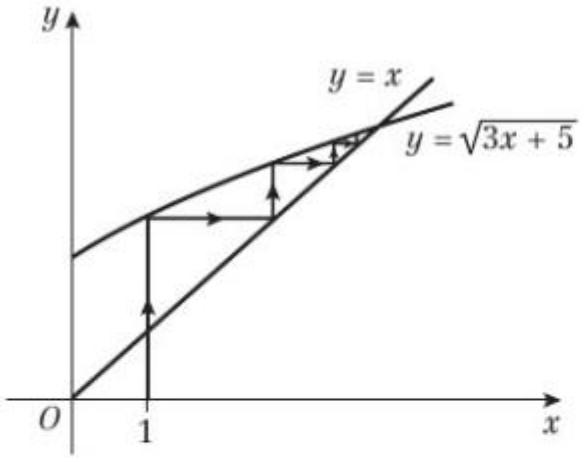
b $g(x) = 0 \Rightarrow x^5 - 5x - 6 = 0$
$\Rightarrow x^5 = 5x + 6 \Rightarrow x = (5x + 6)^{\frac{1}{5}}$
$p = 5, q = 6, r = 5$

c $x_1 = 1.6154$, $x_2 = 1.6971$, $x_3 = 1.7068$

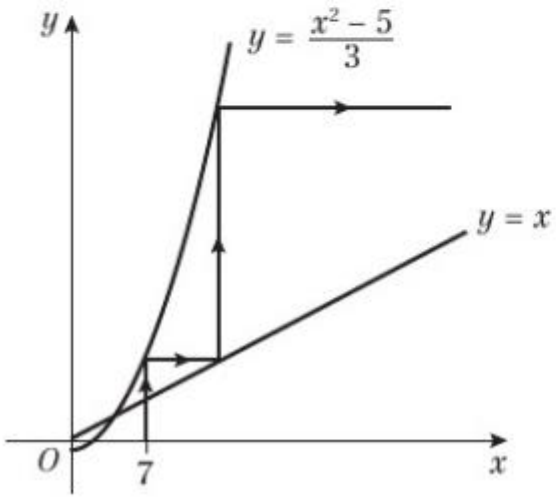
d $g(1.7075) = -0.0229... < 0$, $g(1.7085) = 0.0146... > 0$. The sign change implies there is a root in this interval.

5 **a** $g(x) = 0 \Rightarrow x^2 - 3x - 5 = 0$
$\Rightarrow x^2 = 3x + 5 \Rightarrow x = \sqrt{3x + 5}$

b, c



d

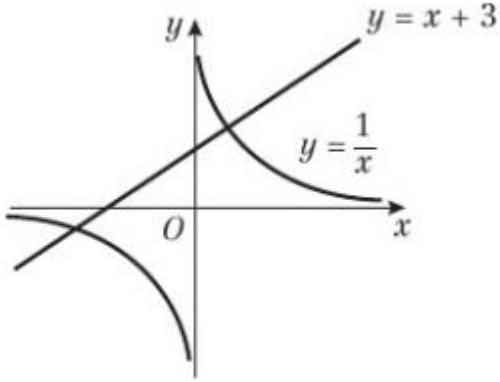


6 **a** $f(1.1) = -0.0648\ldots < 0$, $f(1.15) = 0.0989\ldots > 0$.
The sign change implies there is a root in this interval.

b $5x - 4\sin x - 2 = 0 \Rightarrow 5x = 4\sin x + 2$
$\Rightarrow x = \frac{4}{5}\sin x + \frac{2}{5} \Rightarrow p = \frac{4}{5}, q = \frac{2}{5}$

c $x_1 = 1.113, x_2 = 1.118, x_3 = 1.119, x_4 = 1.120$

7 **a**



b 2

c $\frac{1}{x} = x + 3 \Rightarrow 0 = x + 3 - \frac{1}{x}$, let $f(x) = x + 3 - \frac{1}{x}$
$f(0.30) = -0.0333\ldots < 0$, $f(0.31) = 0.0841\ldots > 0$.
Sign change implies root.

d $\frac{1}{x} = x + 3 \Rightarrow 1 = x^2 + 3x \Rightarrow 0 = x^2 + 3x - 1$

e 0.303

8 **a** $g'(x) = 3x^2 - 14x + 2$ **b** 6.606
c $(x - 1)(x^2 - 6x - 4) \Rightarrow x^2 - 6x - 4 = 0 \Rightarrow x = 3 \pm \sqrt{13}$
d 0.007%

9 **a** $f(0.4) = -0.0285\ldots < 0$, $f(0.5) = 0.2789\ldots > 0$.
Sign change implies root.
b 0.410
c $f(-1.1905) = 0.0069\ldots > 0$,
$f(-1.1895) = -0.0044\ldots < 0$.
Sign change implies root.

10 **a** $e^{0.8x} - \dfrac{1}{3 - 2x} = 0 \Rightarrow (3 - 2x)e^{0.8x} - 1 = 0$
$\Rightarrow (3 - 2x)e^{0.8x} = 1 \Rightarrow 3 - 2x = e^{-0.8x}$
$\Rightarrow 3 - e^{-0.8x} = 2x \Rightarrow x = 1.5 - 0.5e^{-0.8x}$
b $x_1 = 1.32327\ldots$, $x_2 = 1.32653\ldots$, $x_3 = 1.32698\ldots$,
root = 1.327 (3 d.p.)
c $e^{0.8x} - \dfrac{1}{3 - 2x} = 0 \Rightarrow e^{0.8x} = \dfrac{1}{3 - 2x} \Rightarrow 3 - 2x = e^{-0.8x}$
$\Rightarrow -0.8x = \ln(3 - 2x) \Rightarrow x = -1.25\ln(3 - 2x)$
$p = -1.25$
d $x_1 = -2.6302$, $x_2 = -2.6393$, $x_3 = -2.6421$,
root = −2.64 (2 d.p.)

11 **a** $\ln y = x\ln x \Rightarrow \dfrac{1}{y}\dfrac{dy}{dx} = (1)(\ln x) + (x)\left(\dfrac{1}{x}\right)$
$\Rightarrow \dfrac{1}{y}\dfrac{dy}{dx} = \ln x + 1 \Rightarrow \dfrac{dy}{dx} = y(\ln x + 1) = x^x(1 + \ln x)$
b $f(1.4) = -0.3983\ldots < 0$, $f(1.6) = 0.1212\ldots > 0$.
Sign change implies root in the interval.
c $x_1 = 1.5631$ (4 d.p.)
d $f(1.55955) = -0.00017\ldots < 0$,
$f(1.55965) = 0.00011\ldots > 0$. Sign change implies root in the interval.

12 **a** $f(1.3) = -0.18148\ldots$, $f(1.4) = 0.07556\ldots$. There is a sign change in the interval [1.3, 1.4], so there is a root in this interval.
b (0.817, −1.401)
c $x_1 = 0.3424$, $x_2 = 0.3497$, $x_3 = 0.3488$, $x_4 = 0.3489$
d $x_1 = 1.708$
e $f(1.7075) = 0.000435\ldots$, $f(1.7085) = -0.002151\ldots$.
There is a sign change in the interval [1.7075, 1.7085], so there is a root in this interval.

Challenge

a $f(x) = x^6 + x^3 - 7x^2 - x + 3$

$f'(x) = 6x^5 + 3x^2 - 14x - 1$

$f''(x) = 30x^4 + 6x - 14$

$f''(x) = 0 \Rightarrow 15x^4 + 3x - 7 = 0$

i $15x^4 + 3x - 7 = 0 \Rightarrow 3x = 7 - 15x^4 \Rightarrow x = \dfrac{7 - 15x^4}{3}$

ii $15x^4 + 3x - 7 = 0 \Rightarrow 15x^4 + 3x = 7$

$\Rightarrow x(15x^3 + 3) = 7 \Rightarrow x = \dfrac{7}{15x^3 + 3}$

iii $15x^4 + 3x - 7 = 0 \Rightarrow 15x^4 = 7 - 3x$

$\Rightarrow x^4 = \dfrac{7 - 3x}{15} \Rightarrow x = \sqrt[4]{\dfrac{7 - 3x}{15}}$

b Using formula **iii**, root = 0.750 (3 d.p.)

c Formula **iii** gives the positive fourth root, so cannot be used to find a negative root.

d –0.897 (3 d.p.)